University of Pisa

Department of Physics "Enrico Fermi"

From IGC records to measurable trajectories

Methods, results and questions for the supervisors

Matteo Di Sante • 12 September 2026

Chapter 2 • Complete chapter review

The archive contains recorded cross-country flights

	Paragliders	Hang gliders
Catalogue entries	203,343	9,259
Downloaded IGC logs	186,052	6,716
Retained flights	156,406	6,094
Retention	84.1%	90.7%

FFVL Coupe Fédérale de Distance: self-declared flights, heterogeneous recorders, sites and seasons.

Each decoded fix supplies

Time and horizontal position; GNSS and pressure altitude when recorded; receiver validity declaration.

Neither the catalogue nor the retained sample is an unbiased population of all soaring flights.

The analysis population is defined jointly by who records and declares a flight, and by which records our criteria retain.

Current pipeline census. Para catalogue seasons: 1999–2000 to 2025–2026; hang: 2001–2002 to 2025–2026. Downloaded-track coverage differs.

Acquisition preserves the declarations and their recorded tracks

Archived source

Season XML exports retain the original flight lists, metadata and track links. IGC files retain the recorder output.

Derived catalogue

One row per flight links its declared metadata to the local track and download status. Catalogue entries, linked tracks and readable downloads are different counts.

Resumable acquisition checks the recorder header and the presence of fix records. A readable file can still contain physically implausible measurements.

Which missing tracks and declaration choices could change comparisons across seasons or regions?

An IGC fix contains two altitude fields and a validity declaration

Record	Information used
A	Recorder identification
H	Date and descriptive headers
I	Optional extension-field layout
B	UTC time, latitude, longitude, GNSS validity, pressure altitude, GNSS altitude

A syntactically valid coordinate need not be a plausible measurement. The parser checks record structure; cleaning supplies the physical and temporal checks.

A V flag invalidates GNSS altitude here; horizontal position remains provisional until its own checks.

Equipment classes and scoring labels describe different things

Equipment

Paraglider EN/LTF labels are harmonized; CCC, tandem/non-certified and unknown entries remain distinct. Hang-glider classes describe design and control.

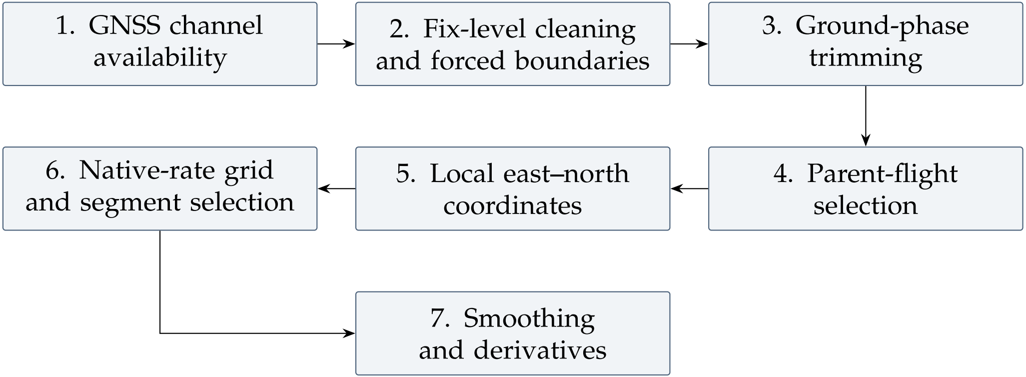
Declared task

Free distance, out-and-return and triangle labels describe scoring categories. They do not label every manoeuvre or establish that a retained segment closes.

The later A/B versus C/D/CCC comparison uses equipment as an experience proxy. Actual pilot skill is not measured by that split.

What independent experience information would allow us to separate pilot choice from equipment?

The processing order determines what each cut acts on



Parent-flight restrictions: recorded duration ≥ 40 min, path ≥ 20 km, altitude range ≥ 600 m, elapsed span ≤ 16 h, plus integrity and reach checks.

These restrictions select a study population. Genuine short flights or nearly level soaring can fail them. Flight-level cuts are not reapplied to every later segment.

GNSS supplies altitude; pressure has a separate witness role

Adopted altitude channel

One channel throughout the population: GNSS presence $\geq$ 95% and range $\geq$ 30 m at flight level.

Missing or invalid GNSS altitude becomes missing z; horizontal coordinates remain subject to their own tests.

The barometer does not substitute for GNSS altitude.

Locally available corroboration

$$B_I = B_{\text{flight}} \prod_{i \in I} b_i.$$

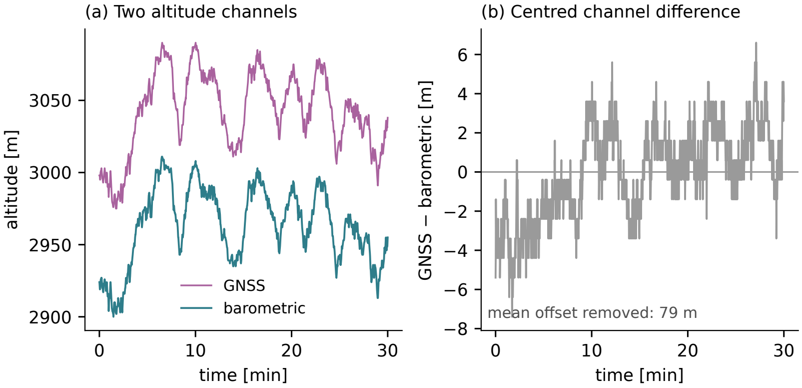
Here $b_i = 1$ for a finite, nonzero pressure reading at fix i, and 0 otherwise.

The candidate interval needs simultaneous position and pressure readings. An incomplete local witness abstains.

How sensitive are witness-dependent decisions to the availability of paired pressure observations?

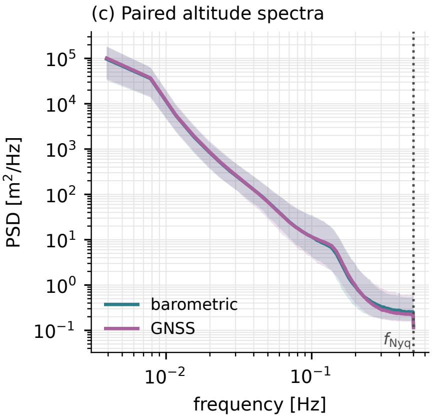
Eligible barometer: flight presence $\geq$ 95%, range $\geq$ 30 m. Eligibility occurs in 72.1% para / 84.7% hang flights attempted by the pipeline.

Recorded altitude traces combine flight motion and channel behaviour



Which common variations plausibly reflect flight motion, and which differences need a sensor explanation?

The paired PSD comparison uses raw valid intervals



Same valid block for both channels; linear resampling and Welch detrending.

No pipeline Savitzky–Golay smoothing enters these spectra.

Shading is the per-frequency 10–90% range across selected flights.

Similar median spectra do not establish identical sensors or identical variation between flights.

Similar typical PSD floors can conceal different upper tails

For each of the 2,008 paired flights, take the median PSD over 0.35–0.50 Hz. The table summarizes this quantity across flights.

Channel	Median	90th percentile	99th percentile
GNSS	2.24×10^{-1}	5.90×10^{-1}	1.38×10^{1}
Barometric	2.61×10^{-1}	5.08×10^{-1}	4.39

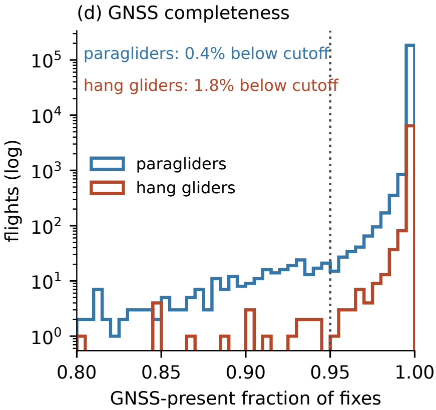
Spectral density in m^2/Hz.

The paired valid-block criteria condition which recorded intervals are measured. The PSD contains motion and recording effects; it is not a pure sensor-noise measurement.

How sensitive are these tails to block length, cadence and logger type?

Current paired-block diagnostic. These floor percentiles differ from the per-frequency shaded range in the preceding plot.

GNSS availability is a separate flight-level selection



The channel gate requires at least 95% usable GNSS altitude and at least 30 m range. The availability census is broader than the selected paired-spectrum population.

Which recorders, periods and flight types are preferentially lost through this requirement?

A common altitude channel does not resolve its vertical datum

The pipeline uses logged GNSS altitude throughout; it does not switch to pressure altitude where GNSS is absent.

$$z_{\text{logged}} = z_{\text{reference}} + b \quad \Rightarrow \quad \Delta z_{\text{logged}} = \Delta z_{\text{reference}} \quad \text{only when } b \text{ is constant.}$$

Recorder conventions and historical compliance remain unresolved. Spatially varying datum corrections and changing sensor biases do not cancel like a constant offset.

Which absolute-height or environmental comparisons need a separate datum audit?

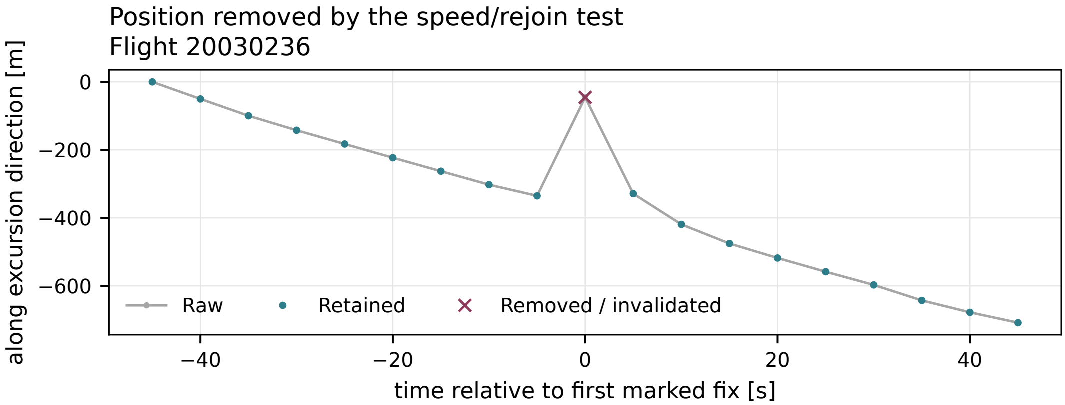
Timestamp and coordinate bookkeeping precede geometric decisions

Duplicate timestamps are consolidated; backward clock steps, impossible coordinates and missing positions are treated explicitly. Exact position repetition also enters the frozen-run witness.

A plausible-looking coordinate can still have the wrong temporal support. Conversely, a sharp turn can disagree with a local median without being a recording defect.

Which structural failures can be detected directly, and which require evidence about the physical trajectory?

A Hampel flag describes a departure; the reach/rejoin rule removes it



Which cases can contaminate the median without defeating the reachability check?

Real flight 20030236; the gate uses geodetic distances. Identifier formula and assumptions.

The identifier's scale is a convention, not a false-alarm probability

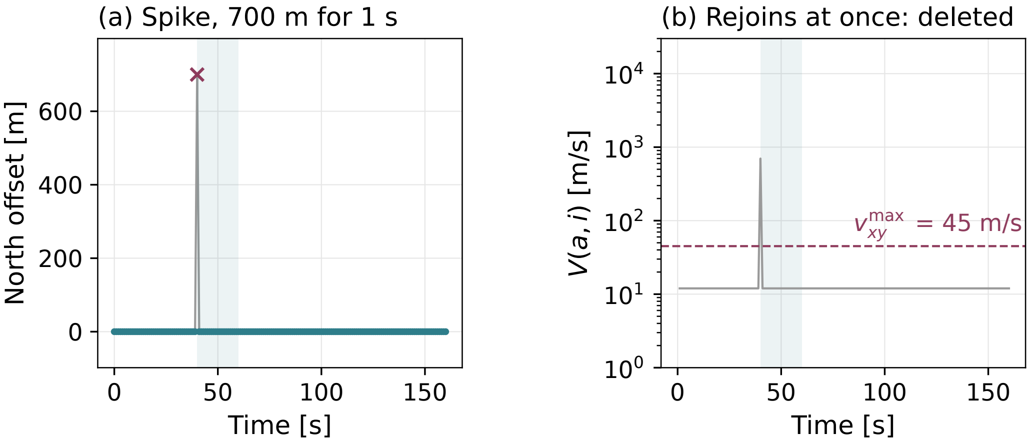
$$r_k = d\big((\varphi_k, \lambda_k), (\widetilde{\varphi}_k, \widetilde{\lambda}_k)\big), \quad \sigma_k = 1.4826\,\mathrm{median}_{|t_j - t_k| \le 20\,\mathrm{s}}\, r_j.$$

$$\text{Flag when } r_k > \max(5\sigma_k, 15\,\mathrm{m}), \quad \text{with at least five fixes in the window.}$$

- Component-wise medians define the local reference position; residuals use horizontal great-circle distance.
- The factor 1.4826 is the one-dimensional Gaussian MAD normalization. For these radial residuals it is an operational scale convention.
- The absolute reach/rejoin gate acts independently of this diagnostic flag.

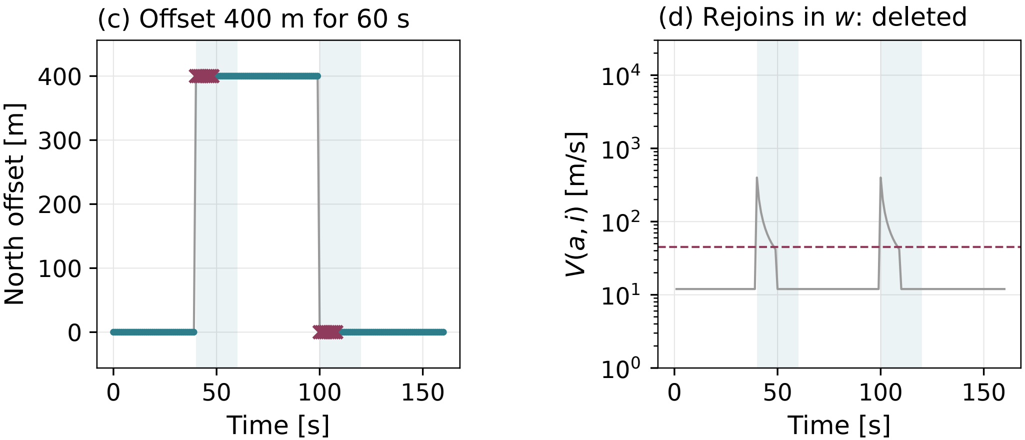
Identifier terminology: Davies & Gather (1993), Example 2.2. Thresholds and GPS removal rules are choices of this analysis.

An isolated unreachable fix has an immediate reachable rejoin



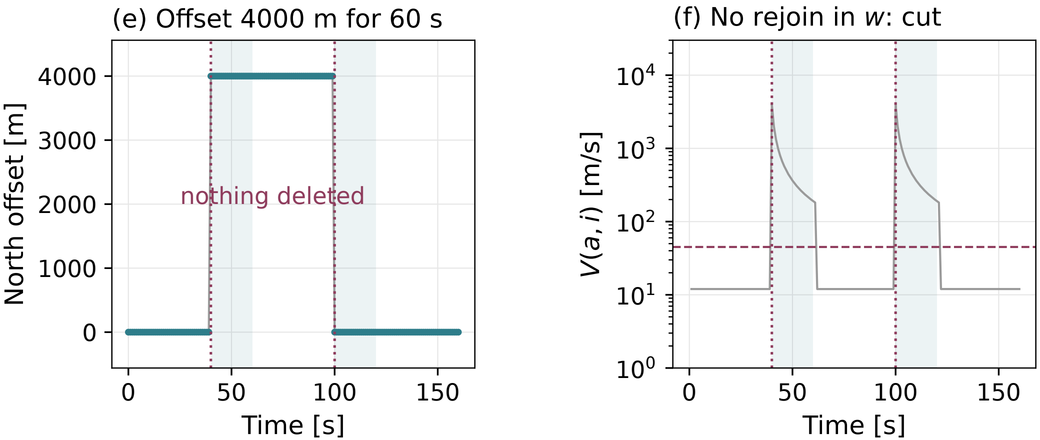
Does deleting the suspect fix restore reachability from the last accepted anchor?

A moderate offset can become reachable within the block horizon



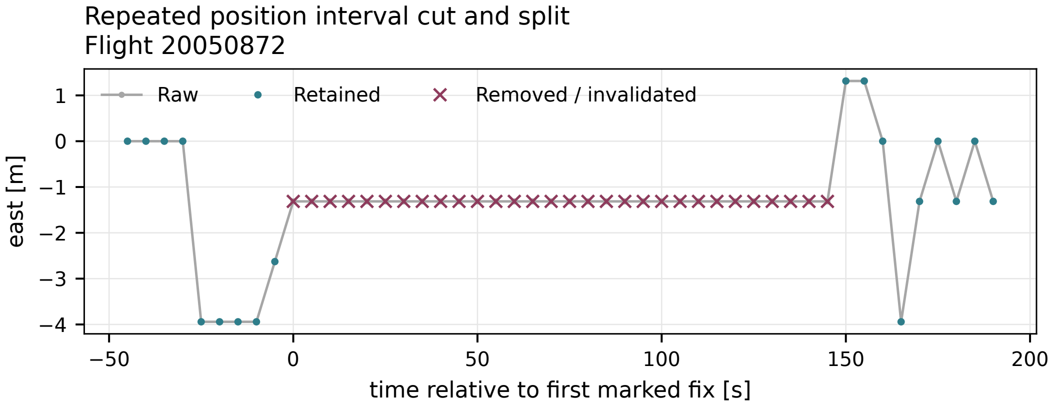
How does keeping the anchor fixed change which early fixes are removed?

Without a reachable rejoin, the scan imposes a boundary



What information remains unresolved after a large reacquisition offset?

A frozen-position candidate is removed and split



How would genuine slow motion differ from this frozen-position signature?

Real flight 20050872; the removed interval remains a gap. Complete witness branches.

The complete frozen-position witness

Require $D_I < 15$ m, $T_I \geq 60$ s and $W_I = 1$. Here D_I is the horizontal bounding-box diagonal.

Let E_I mean exact coordinate repetition; d_I is the fraction with invalid GNSS declaration or missing GNSS altitude. Then

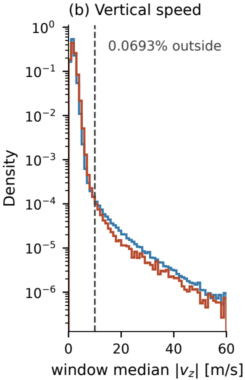
$$W_I = E_I \ \vee \begin{cases} |\Delta b_I| < 5\text{ m}, & B_I = 1, \\ \text{false}, & B_{\text{flight}} = 1,\ B_I = 0, \\ d_I > 1/2, & B_{\text{flight}} = 0. \end{cases}$$

Δb_I is the last-quarter minus first-quarter pressure-altitude median.

For interior ground detection, the pressure test instead uses a detrended 5–95% span and a slope guard, with sustained low speed for at least 10 min. Missing local pressure always causes that guard to abstain.

Small net pressure change can hide an excursion; distinct sensors do not guarantee independent recorder errors. Shorter qualifying slow intervals are flagged and retained.

The vertical-speed threshold is 10 m/s



The threshold lies near the slower-decaying tail and the crossing of the two discipline densities around 9–10 m/s.

Above 10 m/s: 0.069% of window medians in the diagnostic sample.

This is an exceedance fraction, not the fraction of altitudes removed. Both neighbouring decisions still enter the rule.

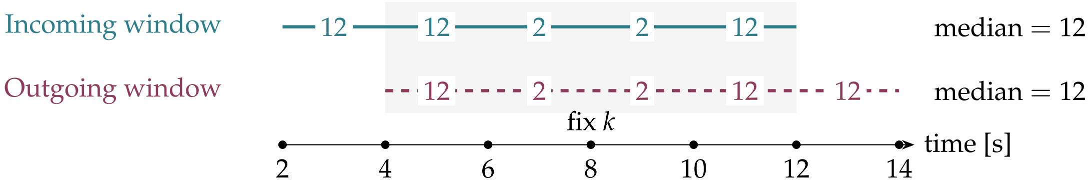
Does changing the threshold materially alter vertical features or phase statistics?

The curve motivates an operational cutoff. It does not establish a physical speed maximum or prove that every larger value is an outlier.

Two window medians determine one altitude decision

With at least five finite steps in a window,

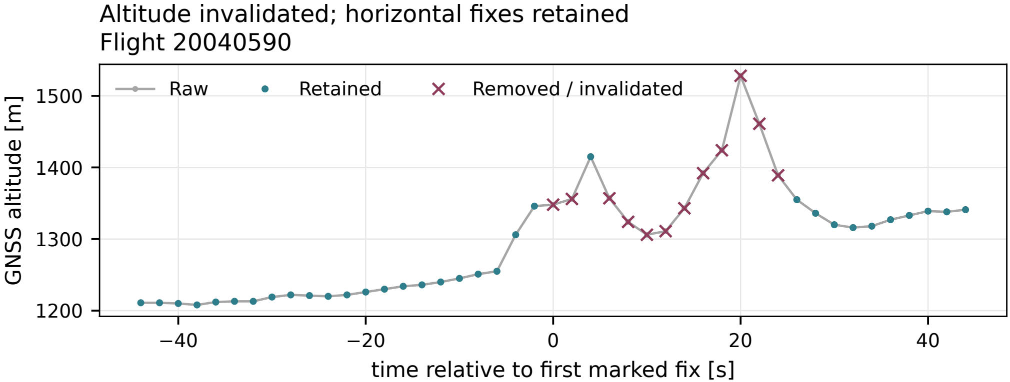
$$u_j = \mathrm{median}_{|m_i - m_j| \le 5\,\mathrm{s}} |v_{z,i}^{\mathrm{obs}}|, \quad m_i = (t_i + t_{i+1})/2; \qquad z_k \to \text{missing if } u_{k-1} > 10 \ \wedge \ u_k > 10.$$



Both medians exceed the threshold although the two steps touching fix k are only 2 m/s. This is a neighbourhood decision.

Schematic with 2-s cadence; speeds in m/s. Below five finite steps, $u_j = |v_{z,j}^{\mathrm{obs}}|$ instead of the median. Equality at 10 m/s does not trigger invalidation.

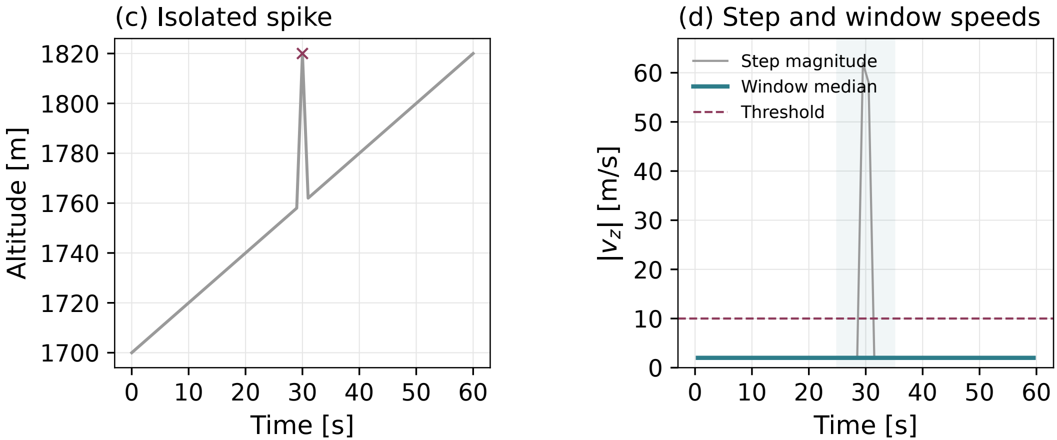
Altitude invalidation precedes the full-trajectory support decision



What evidence could independently distinguish an altitude defect from a genuine manoeuvre?

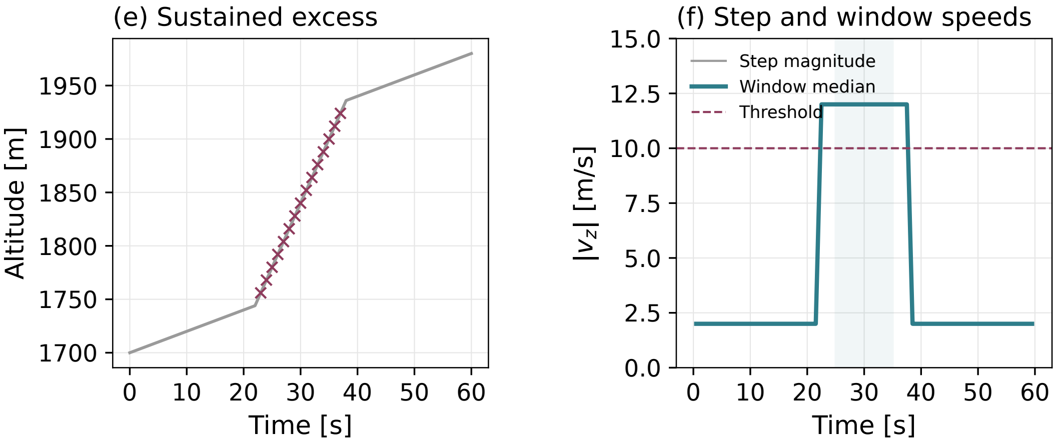
Real paraglider flight 20040590. Synthetic spike and sustained-excess examples isolate what the separate tests do.

An isolated altitude spike can leave a populated-window median low



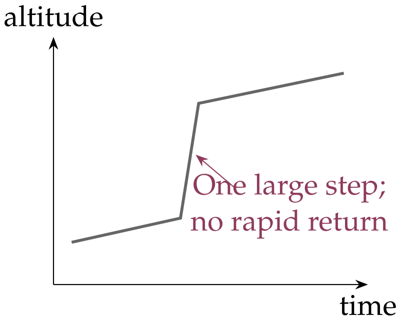
What evidence does the separate opposite-sign bounding-step test add?

Sustained excess raises both neighbouring window statistics



Which plausible physical descents could also satisfy this operational criterion?

An unreturned altitude step remains an unresolved case



A single jump may leave the median low and fail the out-and-back test.

Candidate counters occur in 47,440 para / 1,350 hang retained-flight records, before later trimming and segment selection.

A targeted check found candidate times inside usable phase-feature windows in all eight inspected examples.

Compare exclusion or splitting near candidates before deciding whether an automatic correction is justified.

Schematic. Candidate counts do not count confirmed sensor defects surviving in the final trajectories. No automatic level-shift correction is implemented.

Outer trimming uses a sustained horizontal-speed criterion

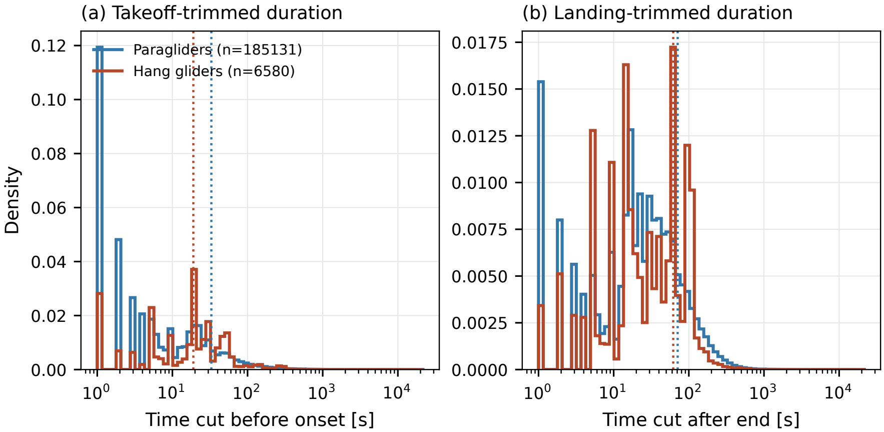
$$v_{xy} > 5\,\mathrm{m\,s^{-1}} \quad \text{for} \quad 30\,\mathrm{s}.$$

Read forward for onset and backward for the final airborne interval. Observed step speeds supply the evidence; slow periods inside the flight do not set these outer boundaries.

Slow airborne motion can be clipped at an end, while sustained fast ground movement can be retained. The persistence time is not an error bound on take-off or landing.

Which boundary cases should be inspected independently of the algorithm's own decision?

Time removed before onset and after landing has a long tail



Do the longest removed intervals reflect continued ground logging or slow flight?

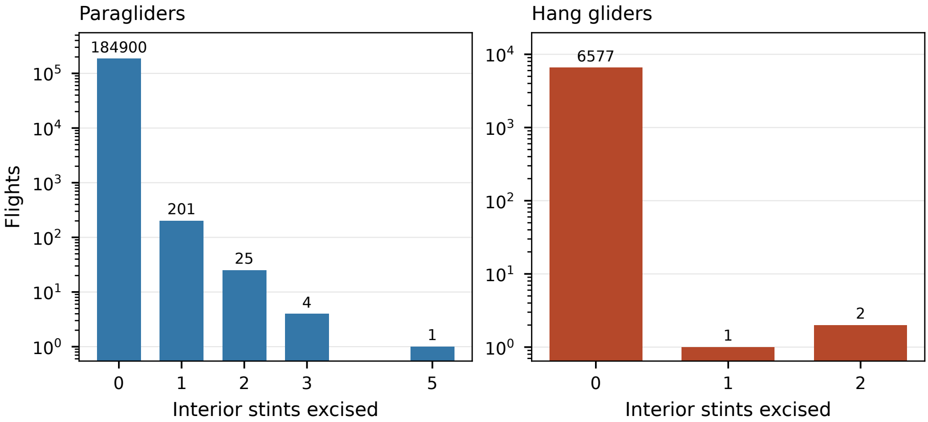
Interior ground intervals require joint speed and barometric evidence

At least ten minutes of low horizontal speed, complete eligible pressure coverage, a small detrended pressure-altitude span, and a bounded fitted vertical trend.

Detrending alone must not erase a genuine steady climb. Without adequate barometry, the detector abstains.

Can ridge or thermal flight meet these conditions, and how would an independent label distinguish it?

Interior excisions affect a small part of the full trimming census



Does a small affected count also imply a small effect on phase statistics?

Flight-level gates define the study population

Quantity	Working restriction
Airborne duration	at least 40 minutes
Observed path length	at least 20 km
Altitude range	at least 600 m
Elapsed span	at most 16 hours

Further integrity and reach checks apply. Segment acceptance follows separately.

Which valid short flights or low-vertical-excursion flights are excluded from the questions we can answer?

Convert geographic coordinates before measuring horizontal motion

$$(\varphi, \lambda, h) \longrightarrow \mathbf{r}_{\mathrm{ECEF}} \longrightarrow \begin{pmatrix} E \\ N \\ U \end{pmatrix} = R(\varphi_0, \lambda_0)\left[\mathbf{r}_{\mathrm{ECEF}} - \mathbf{r}_{\mathrm{ECEF},0}\right].$$

The local orthonormal basis is fixed at the flight origin. East and north are measured in metres, avoiding subtraction of latitude and longitude as if they were Cartesian distances.

On which route lengths would a fixed tangent frame materially affect a geographical comparison?

The local basis uses the origin's geodetic coordinates

$$R(\varphi_0, \lambda_0) = \begin{pmatrix} -\sin\lambda_0 & \cos\lambda_0 & 0 \\ -\sin\varphi_0\cos\lambda_0 & -\sin\varphi_0\sin\lambda_0 & \cos\varphi_0 \\ \cos\varphi_0\cos\lambda_0 & \cos\varphi_0\sin\lambda_0 & \sin\varphi_0 \end{pmatrix}.$$

Subtract the origin in ECEF, then rotate. Geodetic latitude follows the ellipsoid normal; geocentric latitude follows a radius from the Earth's centre.

The IGC latitude already supplies the geodetic angle used in this transformation.

A coordinate convention can create an apparent displacement error before any physical model is fitted.

Rotated Up and the recorded altitude are different coordinates

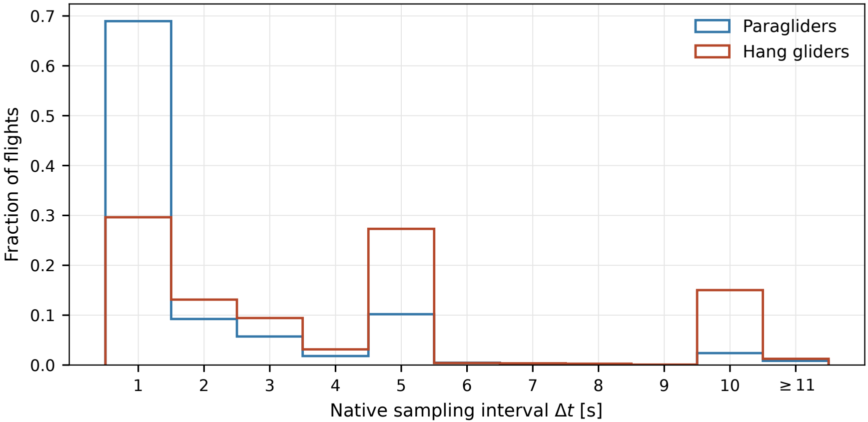
$$(E, N, U) = R\,[\mathbf{r}_{\mathrm{ECEF}} - \mathbf{r}_{\mathrm{ECEF},0}], \qquad z = h_{\mathrm{GNSS,logged}}.$$

The stored vertical channel is logged altitude, not height above the origin's fixed tangent plane. It is not reset to zero at launch.

GNSS altitude supplies a height proxy for the ECEF conversion. Temporary height interpolation used by that conversion does not validate a missing vertical interval; the later gap rule still controls which complete trajectory segments survive.

Which effects depend on an absolute altitude datum, and which cancel in within-segment increments?

Recorder cadence varies across the raw archive



Which native cadences resolve the shortest scales used later?

The same duration bound governs horizontal and vertical support

$$g_{\max} = \min\left(10\Delta t,\ \max(20\,\mathrm{s}, 2\Delta t)\right).$$

Native cadence Δt	1 s	5 s	15 s
Gap bound $g_{\max}$	10 s	20 s	30 s

At or below the bound: interpolate and flag. Above it: split the complete trajectory. For altitude, compare consecutive finite source readings, not the length of a flag run.

How do threshold changes affect both reconstructed motion and which long-lag windows survive?

Long altitude holes split the full trajectory

Flight 20171597 has no admissible vertical support between 6805 s and 7450 s. The 645-s source bracket exceeds the gap bound: one supported segment ends at 6805 s and the next starts at 7450 s.

No fixes from the unsupported interior enter the retained trajectory. Its exclusion is recorded, and the supported pieces undergo the usual segment gates.

Is preserving a complete three-coordinate segment the appropriate trade-off when horizontal position alone remains available?

Raw-flight reprocessing witness for cleaning 2.3.0; the longest remaining altitude-reconstruction flag run in this flight is 1 s.

A segment boundary preserves the flight clock and coordinate origin

Supported pieces keep their parent time and origin. A new segment prevents an increment stencil from joining observations across an unsupported interval.

The observed path length sums within segments. The net displacement between the first and last retained endpoints can still span an unobserved interval.

Which observables depend only on within-segment motion, and which still depend on the separation between recorded pieces?

Segment acceptance requires duration and complete local support

Requirement	Current rule
Segment duration	at least 90 s
Time-interpolated grid fraction	at most 10%
Finite altitude support	each source bracket within $g_{\max}$
Local polynomial	enough samples for its configured window

One parent flight can retain several segments or lose them all. Its original 40-minute and 20-km gates are not reapplied to every segment.

How does the retained population change when the minimum duration or interpolation fraction changes?

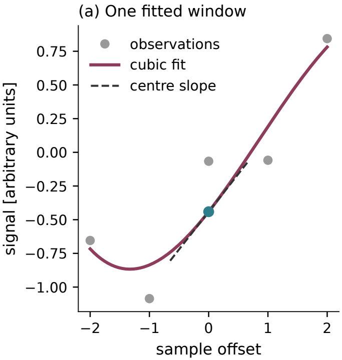
Quality flags describe support; every retained coordinate is smoothed

Flag or boundary	Interpretation
Interpolated	Reconstructed sampling support
Altitude reconstructed	Short missing-altitude support; long holes are split
Edge	Boundary treatment of the local polynomial window
Segment boundary	Increment and feature windows must not cross it

An unflagged interior fix is still part of the filtered trajectory. A sequence of nearest-fix flags is not a direct measurement of finite-source altitude spacing.

Which masks are needed by each observable, and which plausible defects can remain unflagged?

One local polynomial estimates position and derivatives



For offsets $u = -m, \ldots, m$,

$$P_i(u) = \sum_{k=0}^{3} a_k(i) u^k,$$

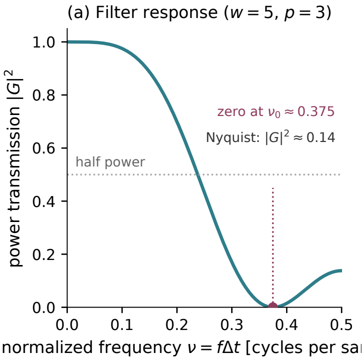
$$\widehat{x}_i = a_0,$$
$$\widehat{v}_i = a_1 / \Delta t,$$
$$\widehat{a}_i = 2a_2 / \Delta t^2.$$

The same fit is applied separately to E, N, z inside each segment.

Smoothing controls derivative noise, but it also changes resolved motion and propagates reconstructed values through its windows.

Synthetic illustration of the implemented Savitzky–Golay fit. Savitzky & Golay (1964).

The nominal 5-s timescale does not mean a 5-s window for every logger



$$w = \max\{\lceil 5\,\mathrm{s}/\Delta t\rceil_{\mathrm{odd}}, 5\}.$$

Five samples span:

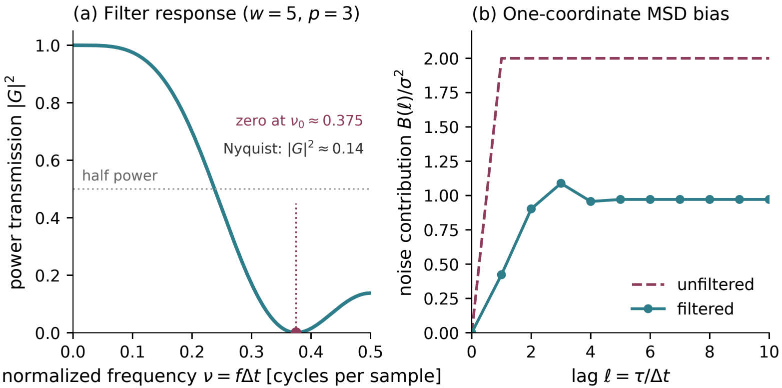
Cadence	Window span
1 s	4 s
3 s	12 s
5 s	20 s

The filter's frequency response depends on cadence.

Quantify the sensitivity of 10-s transport and phase features to cadence and the smoothing window.

The observed spectral knee motivates a working timescale; it is not the exact cutoff of the Savitzky–Golay filter. The response shown is the interior value-filter response.

The value filter changes the contribution of position noise



Left: the interior value-filter response. Right: its MSD contribution for independent white position noise of variance σ^2, in one coordinate.

How would correlated recording errors change this illustrative noise calculation?

Smoothing modifies the measured displacement variance

$$\widehat{x}_j = \sum_k c_k x_{j+k}, \qquad \Delta_\ell \widehat{x}_j = \sum_k c_k \Delta_\ell x_{j+k}.$$

Consequently,

$$\mathbb{E}[(\Delta_\ell \widehat{x})^2] = \sum_{k,m} c_k c_m \, \mathbb{E}[\Delta_\ell x_{j+k} \Delta_\ell x_{j+m}].$$

The effect depends on correlations as well as the filter coefficients. A visually smoother trajectory does not establish that its short-lag MSD is unchanged.

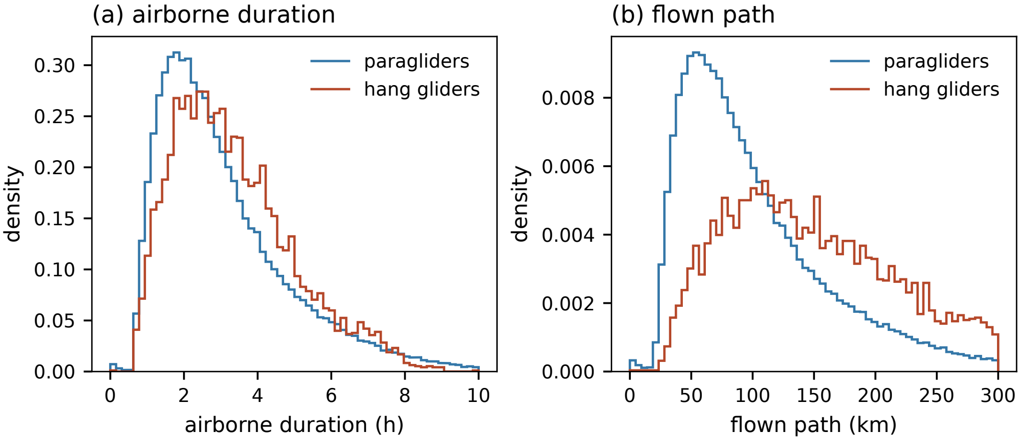
On a common set of flights, which fitted ranges remain stable under cadence and window changes?

The complete first-failure census

Why the flight was dropped	paragliders	hang gliders
unreadable track	27	0
no usable GNSS altitude	697	135
never airborne	197	1
integrity gate	550	53
shorter than the duration cut	2,084	79
longer than the duration cut	3	0
path below the floor	173	1
altitude activity below the floor	4,899	155
out of reach of its own first fix	217	5
no segment survived resampling	20,798	193
no segment carries the smoothing window	1	0
retained	**156,406** (84.1%)	**6,094** (90.7%)

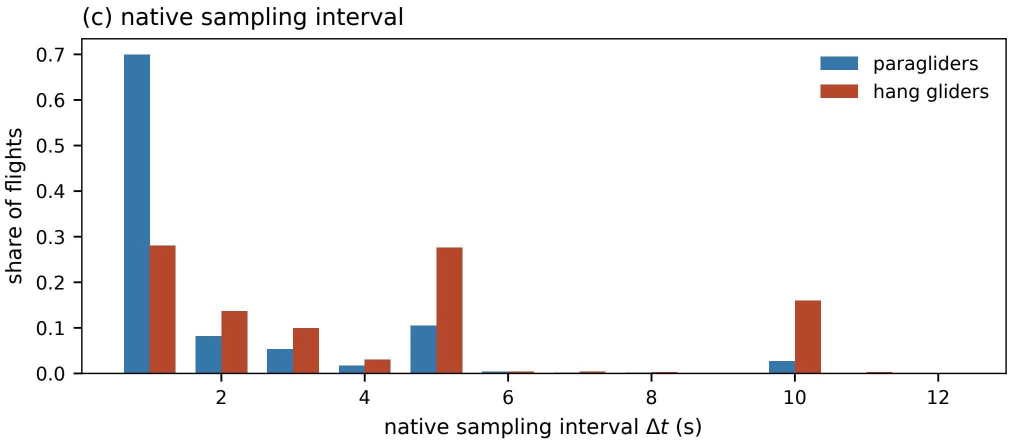
Current clean archive, pipeline version 2.3.0. All numerical entries are imported from the frozen generated table.

Elapsed record span and observed path use different gap conventions



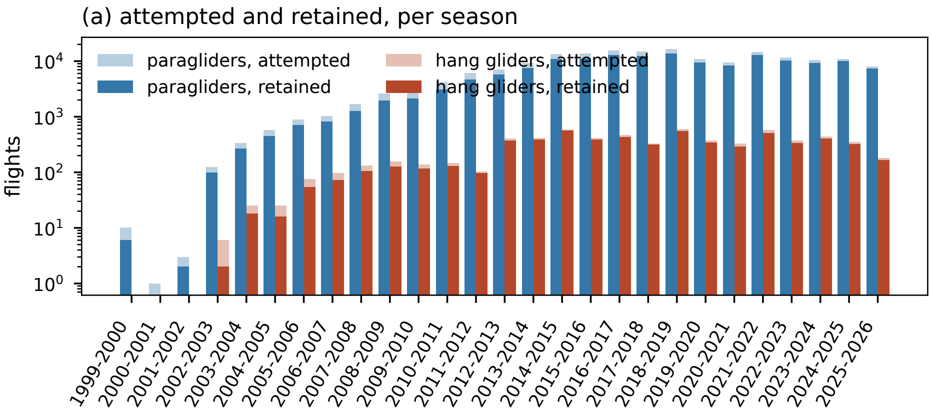
Duration spans the first to last retained fix, including gaps; path sums only within segments.

Retained cadences define the available temporal resolution



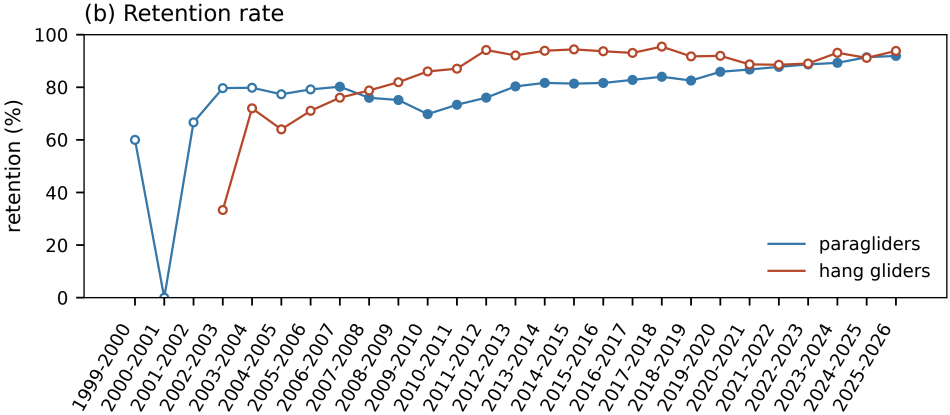
Which results could be sensitive to the changing mixture of recorder intervals?

Participation and available tracks change across seasons



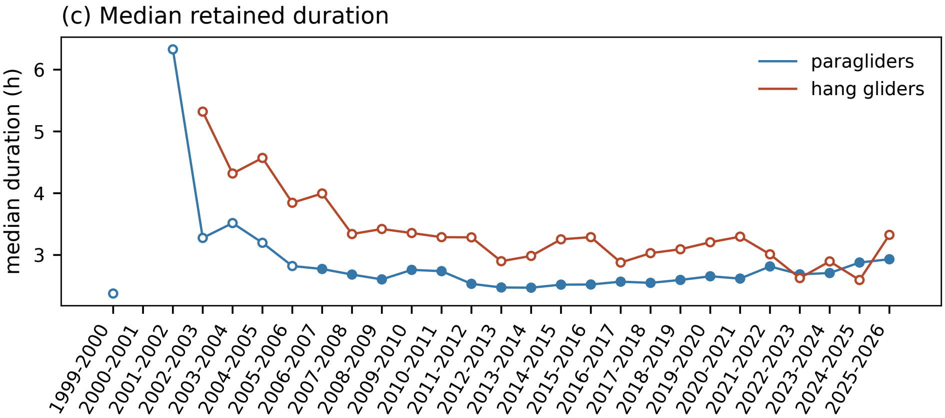
Which temporal comparisons also change equipment, geography or logging technology?

Similar aggregate retention can conceal different selection



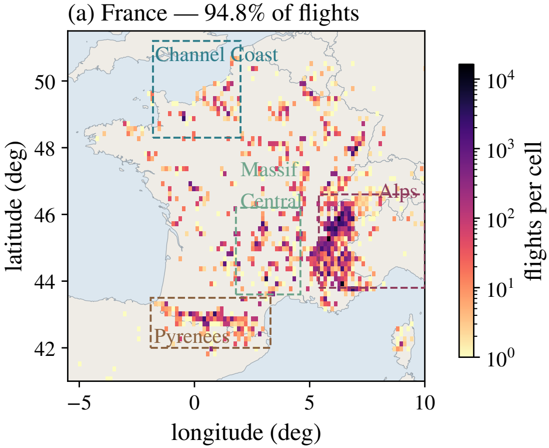
Are cadence and regional retention stable within the larger seasons?

The duration of retained records also changes with season



Would matching season change the apparent equipment or regional contrast?

Geographic coverage is part of the sampling problem

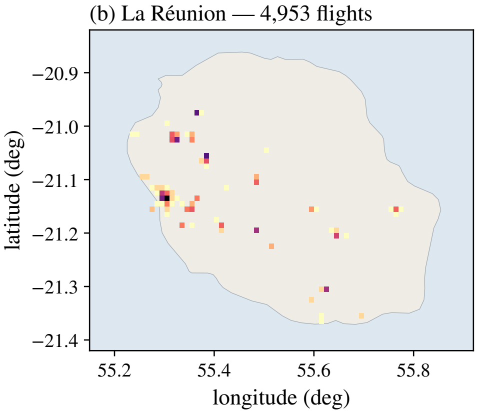


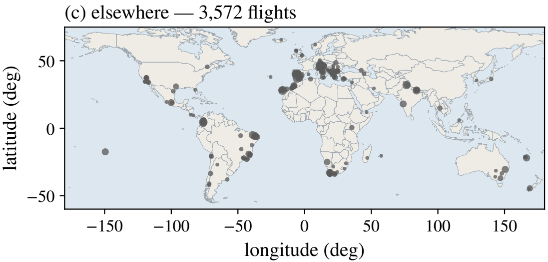
Launch sites and flight characteristics are unevenly represented.

Equipment, season, recording technology and environment can change together.

Directional transport is analysed in Chapter 3; this chapter establishes which observations reach that analysis.

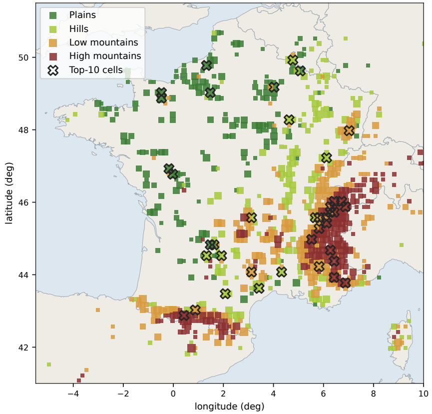
La Réunion and other sites broaden the observed population





Which pooled results combine different climates, seasons and flight environments?

Logged starting altitude does not measure local terrain relief



The map groups first raw altitudes: GNSS where available, with pressure fallback for this exploratory map only.

Its labels (plains, hills, low and high mountains) refer to altitude bands. They do not measure slope, relief or exposure along a route.

Would a terrain model and launch-location check change the environmental strata?

What has been checked, and what still needs empirical validation?

Completed

- Cleaning 2.3.0 and identified numerical products.
- Rule tests, reconstruction masks and trajectory examples.
- Complete structural scan: 1.404 billion retained fixes.
- Reconciliation of numerical reports and manuscript definitions.

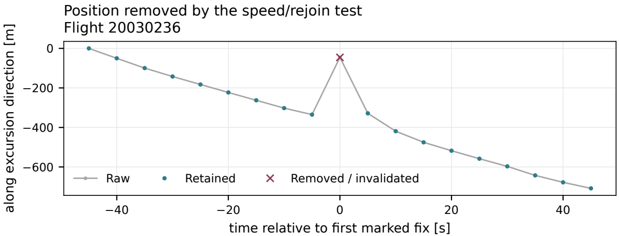
Still open

- Independent detector error rates.
- Observable sensitivity to thresholds, smoothing and cadence.
- Candidate altitude steps and retained suspect intervals.
- GNSS height-datum consistency where it affects conclusions.

Prioritize checks that could change the transport conclusions or the phase features.

Numerical reproducibility and mathematical consistency are necessary. They do not establish physical correctness of every retained fix or every selection rule.

Inspect one of the records together



▸ Open the Python viewer

Paraglider flight **20030236**
Position-spike example around 4410 s after the raw record begins.

Compare the raw and current processed trajectory, then zoom into the suspect interval.

Local macOS link; the companion launcher must be registered. The viewer can open without the SSD; this raw flight needs the connected archive. Offline fallback: the exact frozen example above.

Which cleaning uncertainties should we resolve first?

1. **Vertical reliability.**
   Compare observables with and without neighbourhoods of candidate altitude steps; assess sensitivity to the adopted vertical gap bound.
2. **Sensitivity of the scientific results.**
   Vary the 10-m/s threshold and smoothing/cadence choices on a fixed comparison population; report changes in transport and phase features.
3. **Independent evidence for difficult decisions.**
   Review paired-channel examples and realistic injected defects; distinguish acceptance criteria from measured error rates.

The trajectories are reproducible. Their remaining uncertainties must be quantified where they affect the physics.

Transport plots must retain explicit support and populations

- Retained trajectories: native uniform segments, fixed flight origin and explicit support flags.
- Transport increments: all eligible segments on the stated common grid, with no cross-gap stencil.
- Long-lag comparisons: show changing support and add fixed-population controls.
- Regional interpretations: distinguish observation quality, selection and physical mechanisms.

Which sensitivity analysis would most change confidence in the transport conclusions?

Methods and provenance

Savitzky & Golay (1964)	Polynomial smoothing and derivatives
Fritsch & Butland (1984)	Monotone piecewise cubic interpolation
Davies & Gather (1993)	Hampel identifier

Numerical sources: current Chapter 2 census, paired-channel diagnostic and real cleaning examples. Figure and value snapshots are identified by SHA-256 hashes in the companion manifest.

The cited methods do not establish the validity of the chosen numerical thresholds, the recorder status of individual fixes or the accuracy of downstream phase labels.